Catherine Zhou

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Education

Rice University Houston, TX

B.S. in Computer Science, Minor in Biochemistry/Cell Biology (GPA: 3.93) *Aug. 2023 - May 2027*

- Relevant Courses: EMT-B Course, Discrete Mathematics and Algorithms, Multivariable Calculus, Linear Algebra, Statistics for Data Science, Organic Chemistry 1, Physics E&M, Intermediate Biochemistry, Program Design, Computer Organization

Awards

May 2025 - Rice University Rich Family Endowment Scholarship
Jan. 2025 - Conference on Medicine and Religion Travel Fellowship
Aug. 2024 - NIH/NCI Informatics Technology for Cancer Research (ITCR) Annual Meeting Travel Fellowship
May 2024 - Rice University Goliard Award
June 2024 - Selected as MD Anderson CPRIT-CURE Summer Undergraduate Program Scholar
Jan. 2024 - Rice University Datathon 2024 First Place in Track
May 2023 - Science Olympiad 3rd Team Place at States
June 2022 - American Statistical Association Project Competition National 3rd Place

Experience

Research Assistant Jan. 2021 - Present

Butcher Lab, Stanford University School of Medicine *Palo Alto, CA*

- Developed high-throughput computational pipeline for screening **2M+ protein complexes** using docking and interactions.
- Achieved **80%** reduction in AlphaFold2 run time by optimizing parallel **SLURM** batch scripts and GPU usage on **HPC cluster**.
- Developed self-attention model for ranking complexes using structural/biochemical features, achieving **94.7%** accuracy.
- Identified **two novel** GPCR receptor-ligand pairs that were validated through in vivo assays, contributing to *Nature* publication.
- Developed novel ML pipeline to identify alternative splicing events contributing to melanoma progression in scRNA-seq data.

Research Assistant June 2024 - Present

Ken Chen Lab, MD Anderson Cancer Center *Houston, TX*

- Analyzed scRNA-seq transcriptomic data to profile fibroblast populations in **100+ samples** from head and neck cancer biopsies.
- Building **CNN** with transfer learning to predict binding through interaction, expression, and conservation metrics.

Director Sep. 2023 - Present

Biokind Analytics at Rice *Houston, TX*

- Built partnerships with healthcare nonprofits and directed interdisciplinary teams on community-focused data initiatives.
- Developed visualizations and interactive maps, applying **statistical models** and machine learning to extract key insights.
- Optimized outreach for Candlelighters cancer nonprofit by analyzing **20 years** of data from **5k+ patients** for program growth.

Researcher Jan. 2024 - Present

Institute of Spirituality and Health, Texas Medical Center *Houston, TX*

- Conducted **20+** interviews to identify improvements to spiritual care volunteer program across diverse long-term care settings.
- Applied sentiment analysis, topic modeling, and statistical analysis on **5 years** of data to find patterns in patient interactions

Baylor College of Medicine Patient Discharge Initiative (BCM-PDI) Sep. 2023 - Present

Shift Leader, Presence Committee Head *Houston, TX*

- Serve as patient liaison for underserved populations at the Ben Taub Hospital, facilitating access to healthcare resources.
- Educate patients on discharge instructions, medications, follow-up appointments, and medical/social services.

Site Leader June 2024 - Present

Alternative Spring Break *Houston, TX*

- Design curriculum and lead weeklong spring break trip addressing human trafficking in Houston and New Orleans.
- Foster community engagement through partnerships with local government, organizations, and advocacy groups.

Ambassador Aug. 2024 - Present

Center for Civic Leadership (CCL) *Houston, TX*

- Serve as school-wide ambassador, fostering connections between prospective students and Rice's community-focused programs.
- Increase student involvement in civic engagement projects through monthly events with 30+ attendees and outreach campaigns.

Rice Catalyst Sep. 2023 - Present
Publication Director *Houston, TX*
- Direct copy editing, design, publishing, and distribution processes for Rice's only peer-reviewed undergraduate science journal.

Rice Science Olympiad Alumni Association Sep. 2023 - Present
Outreach Head *Houston, TX*
- Lead marketing for annual Science Olympiad regional tournament hosted by Rice with 500+ middle/high school participants.

Volunteering

Camp Kesem Sep. 2023 - Present
Counselor, AMP Coordinator *Houston, TX*
- Serve as counselor at free annual summer camp for children impacted by a parent's cancer.
- Lead fundraising efforts and design promotional content to raise cancer awareness and support the cause.

Volunteer Jan. 2025 - Present
Monarch Community *Houston, TX*
- Volunteer as spiritual caregiver at residential mental health treatment facility, providing emotional/spiritual support to patients.

Volunteer Sep. 2023 - Present
Lord of the Streets *Houston, TX*
- Provide meals and assist with vision screenings and eye clinic services for unhoused individuals.

Projects

Patient Discharge App | *Python, Django, PostgreSQL, TypeScript, React Native* December 2024 - Present
- Developed a cross-platform mobile app replacing paper-based patient discharge surveys, enabling **70+ volunteers** to scan forms, extract responses via OCR, generate social resource recommendations, and automatically assign and schedule follow-up calls.
- Built a secure backend with **Django** and **PostgreSQL** for storage and management of patient survey data.
- Integrated **REDCap API** to directly import survey data into Ben Taub Hospital data management systems for documentation.

OwlHealth Portal | *Firebase, React, TypeScript, Tailwind CSS, Vite, Node.js* September 2024 - October 2024
- Developed web application providing students with access to Rice University health services through an online portal.
- Integrated **Firebase** for scalable health record storage and appointment scheduling and Cloud Functions for backend logic.
- Implemented **Google OAuth** for authentication and built a real-time chat system using **WebSockets**.

Technical Skills

Languages: Java, Python, C/C++, JavaScript, TypeScript, HTML/CSS, R, SQL, Bash
Frameworks and Libraries: React, Node.js, Vite, pandas, NumPy, Matplotlib, FFmpeg, Whisper, PyTorch, TensorFlow, Folium, GeoPy
Tools and Platforms: Git, Github Actions, Docker, Shell, AWS, Linux, Docker, Kubernetes, Firebase, Selenium
Computational Biology Skills: Seurat, Scanpy, CellRanger, DESeq2, edgeR, rMATS, SUPPA2, GSEA, EnrichR, PANTHER, SAMtools, BWA, STAR, HISAT2, BLAST, Kallisto, Salmon, StringTie, Clustal Omega, MEGA, AutoDock, SwissDock, ClusPro, HADDOCK, Rosetta, ChimeraX, PyMOL, GROMACS, AlphaFold, ColabFold, Boltz, inferCNV, copyKAT, tSpace, scVelo, Monocle, CellPhoneDB, CellChat
Wet Lab Skills: Cell culture, PCR, western blotting, plasmid isolation, CRISPR screening, synthetic organic chemistry, NMR, TLC, mass spectrometry, UV/Vis spectroscopy, IR spectroscopy

Publications and Presentations

"Using machine learning methods to explore significant alternative splicing events in murine melanoma with single cell RNA-seq data", International Conference on Intelligent Systems for Molecular Biology (ISMB) 2022 Madison, WI, USA, July 10-14, 2022 - Posters
Ocón, B. et al. A lymphocyte chemoaffinity axis for lung, non-intestinal mucosae and CNS. Nature (2024). https://doi.org/10.1038/s41586-024-08043-2 - Paper
"Investigating epithelial-mesenchymal transition and endothelial-mesenchymal transition in the tumor microenvironment of HPV+ head and neck cancer", NCI Informatics Technology for Cancer Research (ITCR) Annual Meeting 2024, Indianapolis, IN, USA, September 16-19, 2024 - Poster
"Integration and Efficacy of Volunteer-Based Spiritual Care Services in Hospice and Long-Term Care Settings: A Mixed-Methods Analysis", Conference on Medicien and Religion 2025, Costa Mesa, CA, USA, April 6-8, 2024 - Poster
"Using machine learning methods to explore significant alternative splicing events in murine melanoma with single cell RNA-seq data", International Conference on Intelligent Systems for Molecular Biology (ISMB) 2022, Madison, WI, USA, July 10-14, 2022 - Oral Presentation
Kocalar, S. et al. Hammett linear free-energy relationships in the biocatalytic hydrolysis of para-substituted nitrophenyl benzoate esters. Journal of Emerging Investigators (2021). https://doi.org/10.59720/21-040 - Paper